Romina Aalishah

+1 (443) 7629852
raalish1@jh.edu
LinkedIn Profile
Google Scholar Profile

AREAS OF INTEREST

Machine Learning | Computer Vision | Machine Reasoning | Edge Computing

Education

•Direct Ph.D. in Electrical and Computer Engineering *2024–Present*
Johns Hopkins University, Baltimore, MD GPA: 3.93/4

– M.S. in Electrical and Computer Engineering earned in 2025

•B.Sc. in Electrical Engineering, Minor in Computer Science *2019–2024*
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran GPA: 17.3/20 (3.75/4)

– Minor GPA: 17.5/20 (3.5/4)

•High School Diploma in Mathematics and Physics *2016–2019*
Sampad High School, affiliated with NODET GPA: 19.02/20 (4.0/4.0)

Awards and Honors

- **NeurIPS 2025 Student Volunteer**, San Diego; Selected student volunteer; received complimentary registration.
- **Second Best Paper Award**, AAAI 2025 Spring Symposium; For "GenAI at the Edge: Comprehensive Survey on Empowering Edge Devices."
- **Best Poster Award (People's Choice)**, Johns Hopkins University; For "MambaLiteSR: Image Super-Resolution with Low-Rank Mamba using Knowledge Distillation."
- **DAC 2025 Young Student Fellow** and **Emerging Contributor Award**; Recognized at the 62nd Design Automation Conference.
- **Code-Cup Programming Competition**; Ranked 32/1073 in Data Science and 68/2417 in Algorithms.
- **National Universities Entrance Exam (Iran)**; Ranked 225/164,000 in Mathematics and Physics.

Publications

•MedMambaLite-v2: Shared Selective Scan for Efficient Edge Medical Mamba *2026*
Romina Aalishah, Mozhgan Navardi*, Nithin Kidangazhiath Mana, Tinoosh Mohsenin*

– Submitted to IEEE Transactions on Biomedical Circuits and Systems (TBioCAS), Special Issue: Selected Papers from BioCAS 2025

•Class Unlearning via Depth-Aware Removal of Forget-Specific Directions *2026*
Arman Hatami, Romina Aalishah, Ilya E. Monosov

– CVPR 2026; Workshop on Machine Unlearning for Computer Vision

•EdgeNavMamba: Mamba-Optimized Object Detection for Energy-Efficient Edge Devices *2025*
Romina Aalishah, Mozhgan Navardi, Tinoosh Mohsenin

– 11th IEEE International Conference on Edge Computing and Scalable Cloud (IEEE EdgeCom 2025)

•MedMambaLite: Hardware-Aware Mamba for Medical Image Classification *2025*
Romina Aalishah, Mozhgan Navardi, Tinoosh Mohsenin

– 21st IEEE Biomedical Circuits and Systems Conference (IEEE BioCAS 2025)
– Invited to submit an extended version of this paper to the Special Issue of selected papers from BioCAS 2025 at the IEEE Transactions on Biomedical Circuits and Systems journal.

•GenAI at the Edge: Comprehensive Survey on Empowering Edge Devices *2025*
Mozhgan Navardi, Romina Aalishah, et al.

– AAAI Spring Symposium 2025; Special Session: Generative AI @ Edge
– Won the Second Best Paper Award at this session.

•MambaLiteSR: Image Super-Resolution with Low-Rank Mamba using Knowledge Distillation *2025*
Romina Aalishah, Mozhgan Navardi, Tinoosh Mohsenin

– ISQED Spring Symposium 2025; Special Session: Generative AI in Edge

Research Experiences

•Research Assistant *Feb. 2026 - Present*

Johns Hopkins University, LAMI Baltimore, MD

- Working at the Laboratory of Adaptive and Maladaptive Intelligence (LAMI) under Prof. Ilya Monosov.
- Analyzing interpretability in deep neural networks by studying how internal representations drive model behavior.
- Using knowledge distillation as a controlled framework to compare representations across two networks solving the same task and quantify how compression changes internal computation.
- Evaluating how representation differences between teacher and student predict failure modes and using these insights to guide more interpretable and robust models.

•Research Assistant *Sep. 2024 - Jan. 2026*

Johns Hopkins University, EEHPC Lab Baltimore, MD

- Working at Energy-Efficient High-Performance Computing (EEHPC) laboratory under Prof. Tinoosh Mohsenin.
- Contributing to energy-efficient computing projects focusing on computer vision applications.
- Implementing hardware-aware optimization for compressed deep neural networks to enhance deployment on resource-constrained devices.
- Developing vision-language-action models in robotics projects focused on path planning by leveraging reinforcement learning and brain-inspired deep learning algorithms to enhance autonomous navigation.
- Using IsaacLab, MiniWorld, and Gazebo simulator for better assessment and decreasing the sim-to-real gap.

•Teaching Assistant *Aug. 2025 - Dec. 2025*

Johns Hopkins University Baltimore, MD

- Working as the TA for the course "Design of Advanced Instruments and Systems" under Mr. Sathappan Ramesh.

•Research Assistant *Mar. 2024 - Jul. 2024*

Amirkabir University of Technology(Tehran Polytechnic) Tehran, Iran

- Working at the Laboratory of Computational Intelligence under Prof. Mohammad Bagher Menhaj and Prof. Masoud Shafiee.
- Using reinforcement learning on mobile robots to discover an area.

•Signal Processing Intern *Mar. 2023 - Jun. 2023*

West Pomeranian University of Technology, Faculty of Electrical Engineering Szczecin, Poland

- Work under Prof. Grzegorz Psuj.
- Reviewing resources and comparing the methods related to Magnetic Barkhausen Noise, data processing and knowledge extraction, and deep learning-based robust analysis of magnetic Barkhausen noise using Matlab.

•Back-End Developer *Feb. 2022 - Dec. 2023*

GoldScan Tehran, Iran

- Developing back-end using Django, Django-Rest framework
- Utilizing such technologies as HTML, CSS, and SQL to develop new applications
- Using Postman to test these applications
- Creating Python-Telegram Bot for some parts of projects.

•Electrical Engineering Intern *Sep. 2021 - Mar. 2022*

SYNTECH Technology and Innovation Center at Qazvin Islamic Azad University Qazvin, Iran

- Interface Electronic Circuits to Arduino
- VHDL coding
- Read and analyze papers in the field of Biomedical Engineering (Related to the department projects)

Projects

•Analysis and Assemblage of Comau Smart 5 NJ-165-3.0 Robot *May 2022 - Jun. 2022*

Amirkabir University of Technology

- Analysis, simulation, design, programming, 3D printing, and assemblage of the robot with a scale of 1:15 of the main size.
- Technology Used: SOLIDWORKS, MATLAB, Simulink, Arduino.

•Audit Platform *Dec. 2022 - May 2023*

GoldScan

- Provide the back-end side of an audit platform that automates the auditing process for ledgers, calculating the risk of auditing based on the ledger data and generating worksheets that can be used by auditors to perform the audit.
- Data processing of a given ledger data.
- Provide unit tests to check the performance and use Postman to test the functionality.
- Refactor some parts for better performance.
- Technology Used: Python, Django, Django REST Framework, PostgreSQL, Docker.

Languages

English: Advanced proficiency, TOEFL (iBT) score: 105/120
Persian: Native proficiency
German: Conversational proficiency

Extracurricular Activities

•CISS Staff *Mar. 2025*
Johns Hopkins University Baltimore, MD

- The Conference on Information Sciences and Systems (CISS) is a forum for scientists, engineers, and academics to present their latest research results and developments in multiple areas of Information Sciences and Systems.
- Contributing to the registrations, preparations, and organizing of the conference.

•Chess Team Member *Aug. 2022 - Mar. 2024*
Amirkabir University of Technology (Tehran Polytechnic) Tehran, Iran

- One of the 4 members of university chess team
- The team ranked 5th in the National Universities Rapid Chess Championship 2022.
- The team ranked 7th in the National Universities Blitz Chess Championship 2022.

•Local Committee Member *Jan. 2022 - Feb. 2023*
IAESTE Iran Tehran, Iran

- IAESTE is an international organization exchanging students for technical work experience abroad.
- Socialize with interns from other countries and guide them.